

Ejercicio 1

\$ curl <http://void.ugr.es/publico/noredir/hola.html>

Al realizar el curl de esta dirección nos aparecer por pantalla el código html de esta.

\$ curl -v <http://void.ugr.es/publico/noredir/hola.html>

Al añadirle el modificador -v nos aparece información sobre la dirección:

- Primera línea: indica que estamos solicitando una página con la versión 1.1 de HTTP
- Encabezados: nos indican que estamos realizando una solicitud al host void.ugr.es, con el agente de usuario curl/8.5.0 y que puede aceptar cualquier contenido del tipo MIME.
- Al recibir el servidor el mensaje nos da la siguiente respuesta HTTP
 - Primera línea: indica que esta usando HTTP versión 1.1 y que no ha habido ningún error.
 - Encabezados: nos indican la hora en que se ha enviado la respuesta, que esta usando el servidor Apache/2.4.52, que esta usando un mecanismo de transporte seguro, que la última fecha de modificación del documento fue en febrero de 2024, el tag, el tamaño del fichero en bytes(152), y el tipo MIME text/html.

A parte de toda esta información también nos aparece el código fuente del documento solicitado.

Ejercicio 2

\$ curl -v <http://void.ugr.es/publico/hola.html>

```
* Host void.ugr.es:80 was resolved.
* IPv6: (none)
* IPv4: 150.214.190.100
*   Trying 150.214.190.100:80...
* Connected to void.ugr.es (150.214.190.100) port 80
> GET /publico/hola.html HTTP/1.1
> Host: void.ugr.es
> User-Agent: curl/8.5.0
> Accept: */*
>
< HTTP/1.1 301 Moved Permanently
< Date: Sun, 09 Mar 2025 12:04:42 GMT
< Server: Apache/2.4.52 (Ubuntu)
< Strict-Transport-Security: max-age=0
< Location: https://void.ugr.es/publico/hola.html
< Content-Length: 322
< Content-Type: text/html; charset=iso-8859-1
<
<!DOCTYPE HTML PUBLIC "-//IETF//DTD HTML 2.0//EN">
<html><head>
<title>301 Moved Permanently</title>
</head><body>
<h1>Moved Permanently</h1>
<p>The document has moved <a href="https://void.ugr.es/publico/hola.html">here</a>.</p>
<hr>
<address>Apache/2.4.52 (Ubuntu) Server at void.ugr.es Port 80</address>
</body></html>
* Connection #0 to host void.ugr.es left intact
```

\$ curl -vL <http://void.ugr.es/publico/hola.html>

```
Content-Type: text/html; charset=iso-8859-1

Ignoring the response-body
Connection #0 to host void.ugr.es left intact
Clear auth, redirects to port from 80 to 443
Issue another request to this URL: 'https://void.ugr.es/publico/hola.html'
Host void.ugr.es:443 was resolved.
IPv6: (none)
IPv4: 150.214.190.100
  Trying 150.214.190.100:443...
Connected to void.ugr.es (150.214.190.100) port 443
ALPN: curl offers h2,http/1.1
TLSv1.3 (OUT), TLS handshake, Client hello (1):
  CAfile: /etc/ssl/certs/ca-certificates.crt
  Capath: /etc/ssl/certs
TLSv1.3 (IN), TLS handshake, Server hello (2):
TLSv1.3 (IN), TLS handshake, Encrypted Extensions (8):
TLSv1.3 (IN), TLS handshake, Certificate (11):
TLSv1.3 (IN), TLS handshake, CERT verify (15):
TLSv1.3 (IN), TLS handshake, Finished (20):
TLSv1.3 (OUT), TLS change cipher, Change cipher spec (1):
TLSv1.3 (OUT), TLS handshake, Finished (20):
SSL connection using TLSv1.3 / TLS_AES_256_GCM_SHA384 / X25519 / id-ecPublicKey
ALPN: server accepted http/1.1
Server certificate:
  subject: CN=void.ugr.es
  start date: Mar  8 19:01:44 2025 GMT
  expire date: Jun  6 19:01:43 2025 GMT
  subjectAltName: host "void.ugr.es" matched cert's "void.ugr.es"
  issuer: C=US; O=Let's Encrypt; CN=E6
  SSL certificate verify ok.
  Certificate level 0: Public key type EC/prime256v1 (256/128 Bits/secBits), signed using ecdsa-with-SHA384
  Certificate level 1: Public key type EC/secp384r1 (384/192 Bits/secBits), signed using sha256WithRSAEncryption
  Certificate level 2: Public key type RSA (4096/152 Bits/secBits), signed using sha256WithRSAEncryption
using HTTP/1.x
GET /publico/hola.html HTTP/1.1
Host: void.ugr.es
User-Agent: curl/8.5.0
Accept: */*

TLSv1.3 (IN), TLS handshake, Newsession Ticket (4):
TLSv1.3 (IN), TLS handshake, Newsession Ticket (4):
old SSL session ID is stale, removing
HTTP/1.1 200 OK
Date: Sun, 09 Mar 2025 12:10:16 GMT
```

Ejercicio 3

\$ curl -v -H "Accept-Encoding: gzip" <http://void.ugr.es/publico/noredir/hola.html>

```
* Host void.ugr.es:80 was resolved.
* IPv6: (none)
* IPv4: 150.214.190.100
*   Trying 150.214.190.100:80...
* Connected to void.ugr.es (150.214.190.100) port 80
> GET /publico/noredir/hola.html HTTP/1.1
> Host: void.ugr.es
> User-Agent: curl/8.5.0
> Accept: */*
> Accept-Encoding: gzip
>
< HTTP/1.1 200 OK
< Date: Sun, 09 Mar 2025 12:28:14 GMT
< Server: Apache/2.4.52 (Ubuntu)
< Strict-Transport-Security: max-age=0
< Last-Modified: Tue, 27 Feb 2024 11:08:03 GMT
< ETag: "98-6125b0a4529c4-gzip"
< Accept-Ranges: bytes
< Vary: Accept-Encoding
< Content-Encoding: gzip
< Content-Length: 136
< Content-Type: text/html
<
Warning: Binary output can mess up your terminal. Use "--output -" to tell
Warning: curl to output it to your terminal anyway, or consider "--output
Warning: <FILE>" to save to a file.
* Failure writing output to destination
* Closing connection
```

\$ curl -v -H "Accept-Encoding: gzip" <http://void.ugr.es/publico/noredir/hola.html> -output -

```
* Host void.ugr.es:80 was resolved.
* IPv6: (none)
* IPv4: 150.214.190.100
* Trying 150.214.190.100:80...
* Connected to void.ugr.es (150.214.190.100) port 80
> GET /publico/noredir/hola.html HTTP/1.1
> Host: void.ugr.es
> User-Agent: curl/8.5.0
> Accept: */*
> Accept-Encoding: gzip
>
< HTTP/1.1 200 OK
< Date: Sun, 09 Mar 2025 12:31:08 GMT
< Server: Apache/2.4.52 (Ubuntu)
< Strict-Transport-Security: max-age=0
< Last-Modified: Tue, 27 Feb 2024 11:08:03 GMT
< ETag: "98-6125b0a4529c4-gzip"
< Accept-Ranges: bytes
< Vary: Accept-Encoding
< Content-Encoding: gzip
< Content-Length: 136
< Content-Type: text/html
<
* Connection #0 to host void.ugr.es left intact
9eyJjJ 000;.00D00G00[0F4]
%0DIIfIN00G-N0Bni^J00>D00F00&)?0040000B0R0ZE0"C0R00 0d00E0
```

\$ curl -v -H "Accept-Encoding: gzip" <http://void.ugr.es/publico/noredir/hola.html> -output fichero.html.gz

```
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           Dload  Upload  Total   Spent    Left   Speed
0    0    0    0    0    0     0     0  --:--:-- --:--:-- --:--:--    0* Host void.ugr.
es:80 was resolved.
* IPv6: (none)
* IPv4: 150.214.190.100
* Trying 150.214.190.100:80...
* Connected to void.ugr.es (150.214.190.100) port 80
> GET /publico/noredir/hola.html HTTP/1.1
> Host: void.ugr.es
> User-Agent: curl/8.5.0
> Accept: */*
> Accept-Encoding: gzip
>
< HTTP/1.1 200 OK
< Date: Sun, 09 Mar 2025 12:32:31 GMT
< Server: Apache/2.4.52 (Ubuntu)
< Strict-Transport-Security: max-age=0
< Last-Modified: Tue, 27 Feb 2024 11:08:03 GMT
< ETag: "98-6125b0a4529c4-gzip"
< Accept-Ranges: bytes
< Vary: Accept-Encoding
< Content-Encoding: gzip
< Content-Length: 136
< Content-Type: text/html
<
[ 136 bytes data]
100  136  100  136    0    0   579    0  --:--:-- --:--:-- --:--:--   578
* Connection #0 to host void.ugr.es left intact
```

Al descomprimir el archivo (gunzip fichero.html.gz) podemos ver el código html.

```
<!DOCTYPE html>
<html lang="es">
<head>
  <meta charset="utf-8">
  <title>Hola mundo</title>
</head>
<body>
  <h1>i Hola, mundo !</h1>
</body>
</html>
```

Ejercicio 4

```
alvarogz@agz-ubu24:~$ curl -v http://void.ugr.es/publico/noredir/hola.html
* Host void.ugr.es:80 was resolved.
* IPv6: (none)
* IPv4: 150.214.190.100
*   Trying 150.214.190.100:80...
* Connected to void.ugr.es (150.214.190.100) port 80
> GET /publico/noredir/hola.html HTTP/1.1
> Host: void.ugr.es
> User-Agent: curl/8.5.0
> Accept: */*
>
< HTTP/1.1 200 OK
< Date: Wed, 12 Mar 2025 10:21:32 GMT
< Server: Apache/2.4.52 (Ubuntu)
< Strict-Transport-Security: max-age=0
< Last-Modified: Tue, 27 Feb 2024 11:08:03 GMT
< ETag: "98-6125b0a4529c4"
< Accept-Ranges: bytes
< Content-Length: 152
< Vary: Accept-Encoding
< Content-Type: text/html
<
<!DOCTYPE html>
<html lang="es">
<head>
  <meta charset="utf-8">
  <title>Hola mundo</title>
</head>
<body>
  <h1>¡ Hola, mundo !</h1>
</body>
</html>
* Connection #0 to host void.ugr.es left intact
```

En este caso la última línea nos indica que la conexión no ha sido cerrada, que permanece intacta.

```
alvarogz@agz-ubu24:~$ curl -v -H "Connection: Close" http://void.ugr.es/publico/noredir/hola.html
* Host void.ugr.es:80 was resolved.
* IPv6: (none)
* IPv4: 150.214.190.100
*   Trying 150.214.190.100:80...
* Connected to void.ugr.es (150.214.190.100) port 80
> GET /publico/noredir/hola.html HTTP/1.1
> Host: void.ugr.es
> User-Agent: curl/8.5.0
> Accept: */*
> Connection: Close
>
< HTTP/1.1 200 OK
< Date: Wed, 12 Mar 2025 10:28:00 GMT
< Server: Apache/2.4.52 (Ubuntu)
< Strict-Transport-Security: max-age=0
< Last-Modified: Tue, 27 Feb 2024 11:08:03 GMT
< ETag: "98-6125b0a4529c4"
< Accept-Ranges: bytes
< Content-Length: 152
< Vary: Accept-Encoding
< Connection: close
< Content-Type: text/html
<
<!DOCTYPE html>
<html lang="es">
<head>
  <meta charset="utf-8">
  <title>Hola mundo</title>
</head>
<body>
  <h1>¡ Hola, mundo !</h1>
</body>
</html>
* Closing connection
```

Como podemos ver al usar la extensión -H “Connection: Close” nos indica que la conexión no es persistente, ya que nos indica que ha sido cerrada, otro indicador de que ha sido cerrada es la última línea, Closing connection.


```

alvarogz@agz-ubu24:~$ curl -v -H "Connection: Keep-Alive" http://void.ugr.es/publico/noredir/hola.html
* Host void.ugr.es:80 was resolved.
* IPv6: (none)
* IPv4: 150.214.190.100
* Trying 150.214.190.100:80...
* Connected to void.ugr.es (150.214.190.100) port 80
> GET /publico/noredir/hola.html HTTP/1.1
> Host: void.ugr.es
> User-Agent: curl/8.5.0
> Accept: */*
> Connection: Keep-Alive
>
< HTTP/1.1 200 OK
< Date: Wed, 12 Mar 2025 10:37:31 GMT
< Server: Apache/2.4.52 (Ubuntu)
< Strict-Transport-Security: max-age=0
< Last-Modified: Tue, 27 Feb 2024 11:08:03 GMT
< ETag: "98-6125b0a4529c4"
< Accept-Ranges: bytes
< Content-Length: 152
< Vary: Accept-Encoding
< Keep-Alive: timeout=5, max=100
< Connection: Keep-Alive
< Content-Type: text/html
<
<!DOCTYPE html>
<html lang="es">
<head>
  <meta charset="utf-8">
  <title>Hola mundo</title>
</head>
<body>
  <h1>¡ Hola, mundo !</h1>
</body>
</html>
* Connection #0 to host void.ugr.es left intact

```

En este caso al añadir la extensión -H “Conection: Keep-Alive” se mantiene la conexión persistente, además al igual que en el primer caso la última línea nos indica que la conexión permanece intacta.

Ejercicio 5

```

alvarogz@agz-ubu24:~$ curl -I http://void.ugr.es
HTTP/1.1 301 Moved Permanently
Date: Wed, 12 Mar 2025 10:44:25 GMT
Server: Apache/2.4.52 (Ubuntu)
Strict-Transport-Security: max-age=0
Location: https://void.ugr.es/
Content-Type: text/html; charset=iso-8859-1

```

El servidor de void.ugr.es es Apache y redirige a https de manera permanente.

```

alvarogz@agz-ubu24:~$ curl -I http://www.ugr.es
HTTP/1.1 301 Moved Permanently
Server: nginx
Date: Wed, 12 Mar 2025 10:45:07 GMT
Content-Type: text/html
Content-Length: 162
Connection: keep-alive
Location: https://www.ugr.es/

```

El servidor de www.ugr.es es nginx y redirige a https de manera permanente.

```
alvarogz@agz-ubu24:~$ curl -I http://upm.es
HTTP/1.0 302 Moved Temporarily
Location: https://upm.es/
Server: BigIP
Connection: Keep-Alive
Content-Length: 0
```

El servidor de upm.es es BigIP y redirige a https de forma temporal.

```
alvarogz@agz-ubu24:~$ curl -I http://xkcd.com
HTTP/1.1 301 Moved Permanently
Connection: close
Content-Length: 0
Server: Varnish
Retry-After: 0
Location: https://xkcd.com/
Accept-Ranges: bytes
Date: Wed, 12 Mar 2025 10:50:43 GMT
Via: 1.1 varnish
X-Served-By: cache-mad22070-MAD
X-Cache: HIT
X-Cache-Hits: 0
X-Timer: S1741776643.428153,VS0,VE0
```

El servidor de xkcd.com utiliza varnish como servidor web, indica que cache-mad22070-MAD es el servidor que ha atendido la solicitud y que la respuesta viene de la cache, además redirige a https de manera permanente.

```
alvarogz@agz-ubu24:~$ curl -I http://medium.com
HTTP/1.1 301 Moved Permanently
Date: Wed, 12 Mar 2025 10:51:19 GMT
Content-Type: text/html
Content-Length: 167
Connection: keep-alive
Cache-Control: max-age=3600
Expires: Wed, 12 Mar 2025 11:51:19 GMT
Location: https://medium.com/
X-Content-Type-Options: nosniff
Set-Cookie: _cfuvid=tE.lHUXC1KcIDak6KRpsypWP752AC03WCKuYt2WMg_I-1741776679313-0.0.1.1-604800000; path=/; domain=.medium.com; HttpOnly
Server: cloudflare
CF-RAY: 91f2bc55aa1cec9b-MAD
alt-svc: h3=":443"; ma=86400
```

El servidor es cloudflare y redirige a https de manera permanente.

```
alvarogz@agz-ubu24:~$ curl -I http://medium.com
HTTP/1.1 301 Moved Permanently
Date: Wed, 12 Mar 2025 10:54:08 GMT
Content-Type: text/html
Content-Length: 167
Connection: keep-alive
Cache-Control: max-age=3600
Expires: Wed, 12 Mar 2025 11:54:08 GMT
Location: https://medium.com/
X-Content-Type-Options: nosniff
Set-Cookie: _cfuvid=JKmbgXeRQYLCj24DOVVaDM1KpqP__HLV5A4HAgTKz8o-1741776848427-0.0.1.1-604800000; path=/; domain=.medium.com; HttpOnly
Server: cloudflare
CF-RAY: 91f2c0769963cbf3-MAD
alt-svc: h3=":443"; ma=86400
```

El servidor es cloudfare y redirige a https de manera permanente.

```
alvarogz@agz-ubu24:~$ curl -I http://eldiario.es
HTTP/1.1 301 Moved Permanently
Server: nginx
Date: Wed, 12 Mar 2025 10:54:26 GMT
Content-Type: text/html
Content-Length: 162
Connection: keep-alive
Location: http://www.eldiario.es/
```

El servidor es nginx.

```
alvarogz@agz-ubu24:~$ curl -I http://larazon.es
HTTP/1.1 301 Moved Permanently
Date: Wed, 12 Mar 2025 10:54:56 GMT
Server: Apache/2.4.54 () OpenSSL/1.0.2k-fips
Location: https://www.larazon.es/
Content-Type: text/html; charset=iso-8859-1
```

El servidor es Apache y OpenSSL y redirige a https de manera permanente.

```
alvarogz@agz-ubu24:~$ curl -I http://elpais.es
HTTP/1.1 301 Moved Permanently
Age: 1034
Content-Type: text/html
Date: Wed, 12 Mar 2025 10:38:07 GMT
Location: https://elpais.com/
Server: nginx/1.14.2
Via: 1.1 varnish-v4
X-Cache: HIT
X-Varnish: 145785464 145623403
Connection: keep-alive
```

El servidor es nginx, la solicitud a pasado por por un sistema varnish y redirige a https de forma permanente.

```
alvarogz@agz-ubu24:~$ curl -I http://slack.com
HTTP/1.1 301 Moved Permanently
Server: awselb/2.0
Date: Wed, 12 Mar 2025 10:55:46 GMT
Content-Type: text/html
Content-Length: 134
Connection: keep-alive
Location: https://slack.com:443/
```

El servidor es awselb y redirige a https de forma permanente.

```
alvarogz@agz-ubu24:~$ curl -I http://cni.es
HTTP/1.1 301 Moved Permanently
Date: Wed, 12 Mar 2025 10:56:12 GMT
Server: Apache
X-FRAME-OPTIONS: deny
Set-Cookie: JSESSIONID=AB93E7E5E286D579719793CDA1E1E1F3; Path=/; HttpOnly
Location: https://www.cni.es
Connection: close
Content-Length: 5
Vary: Accept-Encoding
X-XSS-Protection: 1; mode=block
X-Content-Type-Options: nosniff
X-WebKit-CSP: default-src 'self'
X-Permitted-Cross-Domain-Policies: master-only
Strict-Transport-Security: max-age=631138519; includeSubDomains
Referrer-Policy: no-referrer-when-downgrade
Content-Type: text/html; charset=iso-8859-1
```

El servidor es apache y redirige de forma permanente.

Ejercicio 6

```
alvarogz@agz-ubu24:~$ curl -v https://void.ugr.es/publico/procesar.php?nombre=Pepito&password=laclave&enviar=Enviar
[1] 11429
[2] 11430
alvarogz@agz-ubu24:~$ * Host void.ugr.es:443 was resolved.
* IPv6: (none)
* IPv4: 150.214.190.100
* Trying 150.214.190.100:443...
* Connected to void.ugr.es (150.214.190.100) port 443
* ALPN: curl offers h2,http/1.1
* TLSv1.3 (OUT), TLS handshake, Client hello (1):
* CAfile: /etc/ssl/certs/ca-certificates.crt
* CApath: /etc/ssl/certs
* TLSv1.3 (IN), TLS handshake, Server hello (2):
* TLSv1.3 (IN), TLS handshake, Encrypted Extensions (8):
* TLSv1.3 (IN), TLS handshake, Certificate (11):
* TLSv1.3 (IN), TLS handshake, CERT verify (15):
* TLSv1.3 (IN), TLS handshake, Finished (20):
* TLSv1.3 (OUT), TLS change cipher, Change cipher spec (1):
* TLSv1.3 (OUT), TLS handshake, Finished (20):
* SSL connection using TLSv1.3 / TLS_AES_256_GCM_SHA384 / X25519 / id-ecPublicKey
* ALPN: server accepted http/1.1
* Server certificate:
* subject: CN=void.ugr.es
* start date: Mar  8 19:01:44 2025 GMT
* expire date: Jun  6 19:01:43 2025 GMT
* subjectAltName: host "void.ugr.es" matched cert's "void.ugr.es"
* issuer: C=US; O=Let's Encrypt; CN=E6
* SSL certificate verify ok.
* Certificate level 0: Public key type EC/prime256v1 (256/128 Bits/secBits), signed using ecdsa-with-SHA384
* Certificate level 1: Public key type EC/secp384r1 (384/192 Bits/secBits), signed using sha256WithRSAEncryption
* Certificate level 2: Public key type RSA (4096/152 Bits/secBits), signed using sha256WithRSAEncryption
* using HTTP/1.x
> GET /publico/procesar.php?nombre=Pepito HTTP/1.1
> Host: void.ugr.es
> User-Agent: curl/8.5.0
> Accept: */*
>
* TLSv1.3 (IN), TLS handshake, NewSession Ticket (4):
* TLSv1.3 (IN), TLS handshake, NewSession Ticket (4):
* old SSL session ID is stale, removing
< HTTP/1.1 200 OK
< Date: Wed, 12 Mar 2025 11:38:28 GMT
< Server: Apache/2.4.52 (Ubuntu)
< Strict-Transport-Security: max-age=0
< Vary: Accept-Encoding
< Content-Length: 341
< Content-Type: text/html; charset=UTF-8
<
<!DOCTYPE html>
<html>
<head>
```

< > C void.ugr.es/publico/procesar.php?nombre=Pepito&password=laclave&enviar=Enviar

Variables recibidas

Desde \$_GET

- nombre = **Pepito**
- password = **laclave**
- enviar = **Enviar**

Desde \$_POST

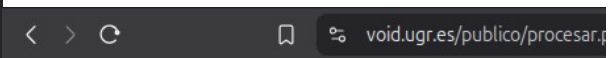
Get envía los datos en la URL y los podemos modificar y además nos permite realizarlo también desde la terminal con curl.

Ejercicio 7

curl -d "nombre=Pepito&password=laclave&enviar=Enviar" -v <https://void.ugr.es/publico/procesar.php>

Este comando envía los datos como cuerpo de la solicitud mientras que al utilizarlo con la extensión `-trace-ascii` nos muestra como los datos están siendo realmente enviados.

```
alvarog@agz-ubu24:~$ curl -d "nombre=Pepito&password=laclave&enviar=Enviar" --trace-ascii - https://void.ugr.es/publico/procesar.php
== Info: Host void.ugr.es:443 was resolved.
== Info: IPv6: (none)
== Info: IPv4: 150.214.190.100
== Info: Trying 150.214.190.100:443...
== Info: Connected to void.ugr.es (150.214.190.100) port 443
== Info: ALPN: curl offers h2,http/1.1
=> Send SSL data, 5 bytes (0x5)
0000: .....
== Info: TLSv1.3 (OUT), TLS handshake, Client hello (1):
=> Send SSL data, 512 bytes (0x200)
0000: .....[G.....[.H.F.[.....%9....].B...t.O....//...NeM...<
0000: ..0....+.....+//...$(.k.#'.g....9....3.....>
0000: <.2/....+.....void.ugr.es.....
0000: .....h2.http/1.1.....1.....*(.....
0100: .....+.....3.8.$...9mL.W.....b3.j..qV..]
0140: [...]0.....
0100: .....
0100: .....
== Info: CAfile: /etc/ssl/certs/ca-certificates.crt
== Info: Capath: /etc/ssl/certs
<= Recv SSL data, 5 bytes (0x5)
0000: .....
== Info: TLSv1.3 (IN), TLS handshake, Server hello (2):
<= Recv SSL data, 122 bytes (0x7a)
0000: ..v..= e)N+0....7.5w.W....."d&.tg...t.O....//...NeM...<
0040: ..0....+.....3.$...k.....F..U..R...[3DKE]....6...4
<= Recv SSL data, 5 bytes (0x5)
0000: .....
<= Recv SSL data, 5 bytes (0x5)
0000: .....
<= Recv SSL data, 1 bytes (0x1)
0000: .
== Info: TLSv1.3 (IN), TLS handshake, Encrypted Extensions (8):
<= Recv SSL data, 25 bytes (0x19)
0000: .....http/1.1
<= Recv SSL data, 5 bytes (0x5)
0000: .....
<= Recv SSL data, 1 bytes (0x1)
0000: .
== Info: TLSv1.3 (IN), TLS handshake, Certificate (11):
<= Recv SSL data, 2024 bytes (0x7e8)
0000: .....[0..w0.....p.#...[.L.N...0...".H...021.0...
0040: U...u51.0...U...let's Encrypt1.0...U...E6...250308190144Z.2
0080: S0606190143Z0.1.0...U...void.ugr.es090...".H...+..B...
00c0: H..t)xI9.....U...X.....[ha.ZD...18E...
0100: ..0..0..U.....0..U.W..0...+.....0..U...
0140: ..0..0..U...U..K.57..x[Y.....0..U.#.0...F...Qh...BH.#
0180: X..01.....0...http://66-0-10encr.org0...>
01c0: B...http://66-0-10encr.org0...U...0...void.ugr.es0...>
```



Variables recibidas

Desde \$_GET

Desde \$_POST

- nombre = **Pepito**
- password = **laclave**
- enviar = **Enviar**

Ejercicio 8

Para que al hacer el curl funcione de esta manera lo que he hecho ha sido cambiar el formulario poniendo `<form action="https://void.ugr.es/~miusuario/procesar.php" method="POST">`, así de esta manera el action apunta hacia el.

```
alvarog@agz-ubu24:~/Escritorio/TW/practicas/pr2$ curl --user usuario:clave -d "nombre=pepito&password=laclave&enviar=Enviar" -v https://void.ugr.es/~USUARIO/procesar.php
* Host void.ugr.es:443 was resolved.
* IPv6: (none)
* IPv4: 150.214.190.100
*   Trying 150.214.190.100:443...
* Connected to void.ugr.es (150.214.190.100) port 443
* ALPN: curl offers h2,http/1.1
* TLSv1.3 (OUT), TLS handshake, Client hello (1):
* CAfile: /etc/ssl/certs/ca-certificates.crt
* CApath: /etc/ssl/certs
* TLSv1.3 (IN), TLS handshake, Server hello (2):
* TLSv1.3 (IN), TLS handshake, Encrypted Extensions (8):
* TLSv1.3 (IN), TLS handshake, Certificate (11):
* TLSv1.3 (IN), TLS handshake, CERT verify (15):
* TLSv1.3 (IN), TLS handshake, Finished (20):
* TLSv1.3 (OUT), TLS change cipher, Change cipher spec (1):
* TLSv1.3 (OUT), TLS handshake, Finished (20):
* SSL connection using TLSv1.3 / TLS_AES_256_GCM_SHA384 / X25519 / id-ecPublicKey
* ALPN: server accepted http/1.1
* Server certificate:
*  subject: CN=void.ugr.es
*   start date: Mar  8 19:01:44 2025 GMT
*   expire date: Jun  6 19:01:43 2025 GMT
*   subjectAltName: host "void.ugr.es" matched cert's "void.ugr.es"
*   issuer: C=US; O=Let's Encrypt; CN=E6
*   SSL certificate verify ok.
*   Certificate level 0: Public key type EC/prime256v1 (256/128 Bits/secBits), signed using ecdsa-with-SHA384
*   Certificate level 1: Public key type EC/secp384r1 (384/192 Bits/secBits), signed using sha256withRSABitEncryption
*   Certificate level 2: Public key type RSA (4096/152 Bits/secBits), signed using sha256withRSABitEncryption
* using HTTP/1.x
* Server auth using Basic with user 'usuario'
> POST /~USUARIO/procesar.php HTTP/1.1
> Host: void.ugr.es
> Authorization: Basic dXN1YXJpb2pjbGFGZ2ZQ==
> User-Agent: curl/8.5.0
> Accept: */*
> Content-Length: 44
> Content-Type: application/x-www-form-urlencoded
>
* TLSv1.3 (IN), TLS handshake, Newsession Ticket (4):
* TLSv1.3 (IN), TLS handshake, Newsession Ticket (4):
* old SSL session ID is stale, removing
< HTTP/1.1 401 Unauthorized
< Date: Wed, 12 Mar 2025 12:23:00 GMT
< Server: Apache/2.4.52 (Ubuntu)
< Strict-Transport-Security: max-age=0
< Authentication problem. Ignoring this.
< WWW-Authenticate: Basic realm="Acceso restringido a usuario de TW"
< Content-Length: 459
< Content-Type: text/html; charset=iso-8859-1
<
<!DOCTYPE HTML PUBLIC "-//IETF//DTD HTML 2.0//EN">
<html><head>
<title>401 Unauthorized</title>
</head><body>
<h1>Unauthorized</h1>
<p>This server could not verify that you
are authorized to access the document
requested. Either you supplied the wrong
credentials (e.g., bad password), or your
```

Ejercicio 9

void.ugr.es/publico/procesarfi.php

Variables recibidas

Desde \$_GET

Desde \$_POST

- nombre = **Pepito**
- password = **laclave**
- enviar = **Enviar**

Ficheros recibidos:

- imagen =
Array
(
[name] => elipse.png
[full_path] => elipse.png
[type] => image/png
[tmp_name] => /tmp/phpFVE4vD
[error] => 0
[size] => 518
)

```
alvarogz@agz-ubu24:~/Escritorio/TW/practicas/pr2$ curl -F "nombre=Pepito&password=laclave&enviar=Enviar" -F "imagen=@elipse.png" https://void.ugr.es/publico/procesarfi.php > resultado.html
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           Dload  Upload  Total   Spent    Left   Speed
100 2192  100 1335  100 857    1954   1254  --:--:-- --:--:-- --:--:-- 3204
alvarogz@agz-ubu24:~/Escritorio/TW/practicas/pr2$ open resultado.html
alvarogz@agz-ubu24:~/Escritorio/TW/practicas/pr2$ Se está abriendo en una sesión de navegador existente.
```

Archivo /home/alvarogz/Escritorio/TW/practicas/pr2/resultado.html

Variables recibidas

Desde \$_GET

Desde \$_POST

- nombre = **Pepito&password=laclave&enviar=Enviar**

Ficheros recibidos:

- imagen =
Array
(
[name] => elipse.png
[full_path] => elipse.png
[type] => image/png
[tmp_name] => /tmp/phpSoCcWR
[error] => 0
[size] => 518
)

Ejercicio 10

Buscar libros

[Libros](#)[Autores](#)[Buscar dentro](#)[Temas](#)[Listas](#)[Búsqueda avanzada](#)

☒ Todo ☐ Libros electrónicos

6 coincidencias

```
alvarogz@agz-ubu24:~/Escritorio/TW/practicas/pr2$ curl -v http://openlibrary.org/search.json?q=Herb+Sutter
* Host openlibrary.org:80 was resolved.
* IPv6: (none)
* IPv4: 207.241.234.205
* Trying 207.241.234.205:80...
* Connected to openlibrary.org (207.241.234.205) port 80
> GET /search.json?q=Herb+Sutter HTTP/1.1
> Host: openlibrary.org
> User-Agent: curl/8.5.0
> Accept: */*
>
< HTTP/1.1 200 OK
< Server: nginx/1.26.3
< Date: Wed, 12 Mar 2025 12:16:30 GMT
< Content-Type: application/json
< Transfer-Encoding: chunked
< Connection: keep-alive
< access-control-allow-origin: *
< access-control-allow-method: GET, OPTIONS
< access-control-max-age: 86400
< x-ol-stats: "SR 1 0.015 TT 0 0.017"
< Referrer-Policy: no-referrer-when-downgrade
<
{
  "numFound": 9,
  "start": 0,
  "numFoundExact": true,
  "num_found": 9,
  "documentation_url": "https://openlibrary.org/dev/docs/api/search",
  "q": "Herb Sutter",
  "offset": null,
  "docs": [
    {
      "author_key": [
        "OL1390780A"
      ],
      "author_name": [
        "Herb Sutter"
      ],
      "cover_edition_key": "OL3297340M",
      "cover_i": 192375,
      "edition_count": 3,
      "first_publish_year": 2004,
      "has_fulltext": false,
      "ia": [
        "isbn_0321113586_958"
      ],
      "key": "/works/OL5718566W",
      "language": [
        "eng"
      ],
      "public_scan_b": false,
      "title": "C++ coding standards"
    }
  ],
}
```